SIMATS ENGINEERING
ASSESSMENT TOOL 1

NAME: P. GAYATHRI **REG.NO:** 192425277
COURSE CODE: Data Warehousing and Data Mining **COURSE NAME:** CSA1603

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50

QUESTIONS: 5 **Total Marks:50**

Annexure A

Experiments

Criteria	Understanding of Concepts ***(2.5)***	Experimental Design ***(3)***	Use of Tools ***(2)***	Analysis of Results ***(1.5)***	Documentation ***(1)***	***Total(10)***
Weightage	*25%*	30%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I P. GAYATHRI/192425277_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

P. GAYATHRI
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

1.Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

ANS:

Covariance and Correlation

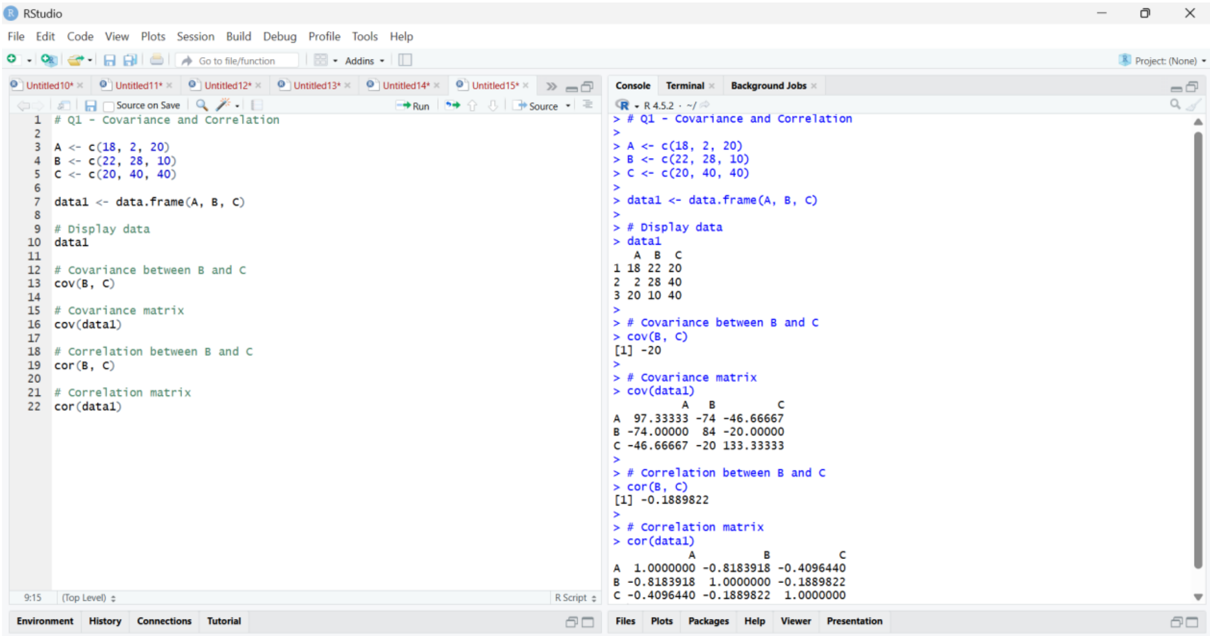
The question gives the following preferences for photographs A, B and C and asks for covariance and correlation using cov() and cor().

Program R Code

```
# Q1 - Covariance and Correlation

A <- c(18, 2, 20)

B <- c(22, 28, 10)

C <- c(20, 40, 40)

data1 <- data.frame(A, B, C)

# Display data

data1
```

```
# Covariance between B and C

cov(B, C)

# Covariance matrix

cov(data1)

# Correlation between B and C

cor(B, C)

# Correlation matrix

cor(data1)
```

Output:



Result

The sample covariance between B and C is **-20**, and the sample correlation is approximately **-0.189**. The covariance and correlation matrices were successfully calculated.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10

Player ID	Player Name	Points Scored
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

ANS:

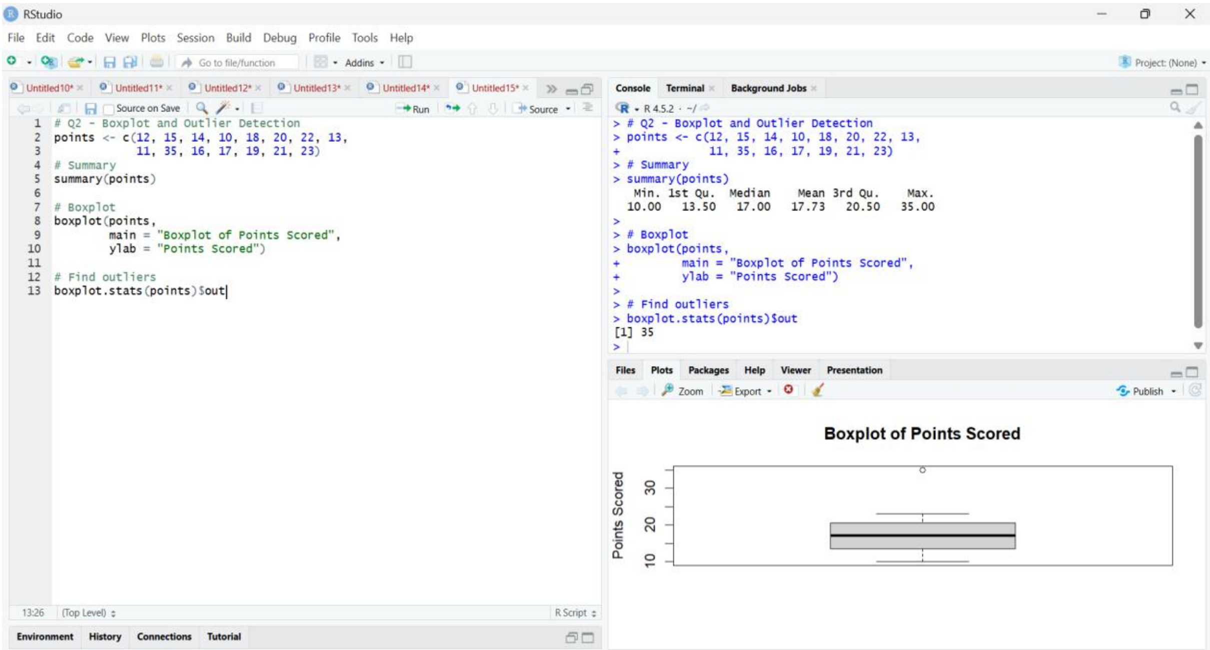
Boxplot and Outlier Detection

The question gives points scored by 15 tennis players and asks how a coach can use a boxplot to identify outliers.

R Code:

```
# Q2 - Boxplot and Outlier Detection

points <- c(12, 15, 14, 10, 18, 20, 22, 13,

            11, 35, 16, 17, 19, 21, 23)

# Summary

summary(points)

# Boxplot

boxplot(points,

        main = "Boxplot of Points Scored",

        ylab = "Points Scored")

# Find outliers

boxplot.stats(points)$out
```

Output:



Result:

Player J, who scored **35 points**, is an outlier.

The boxplot shows that most players scored between approximately **10 and 23 points**, while 35 is unusually high.

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**. (10 Marks)

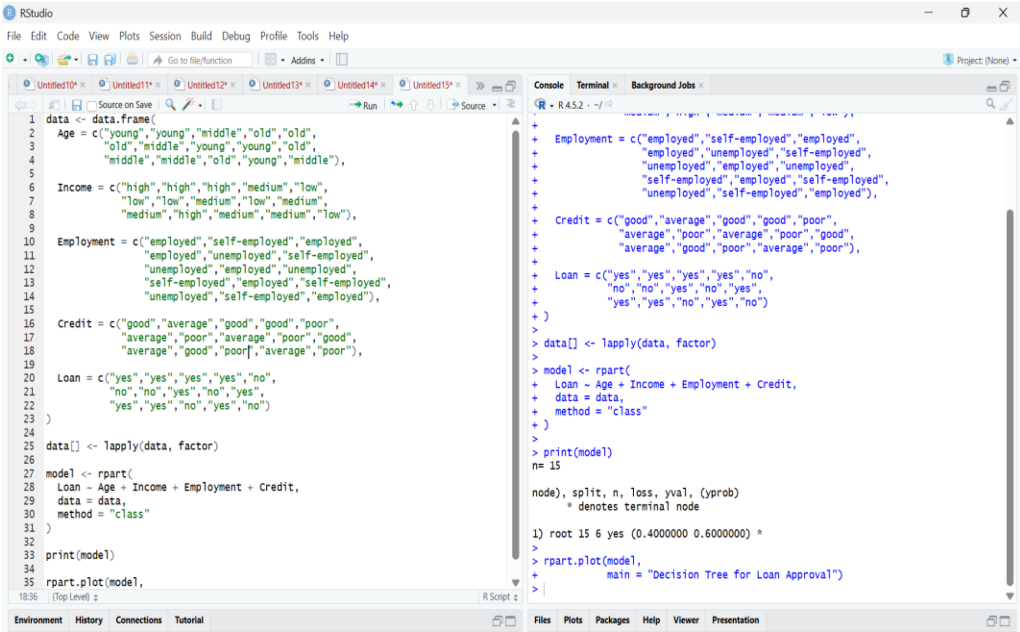
Consider the Data Set
young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes

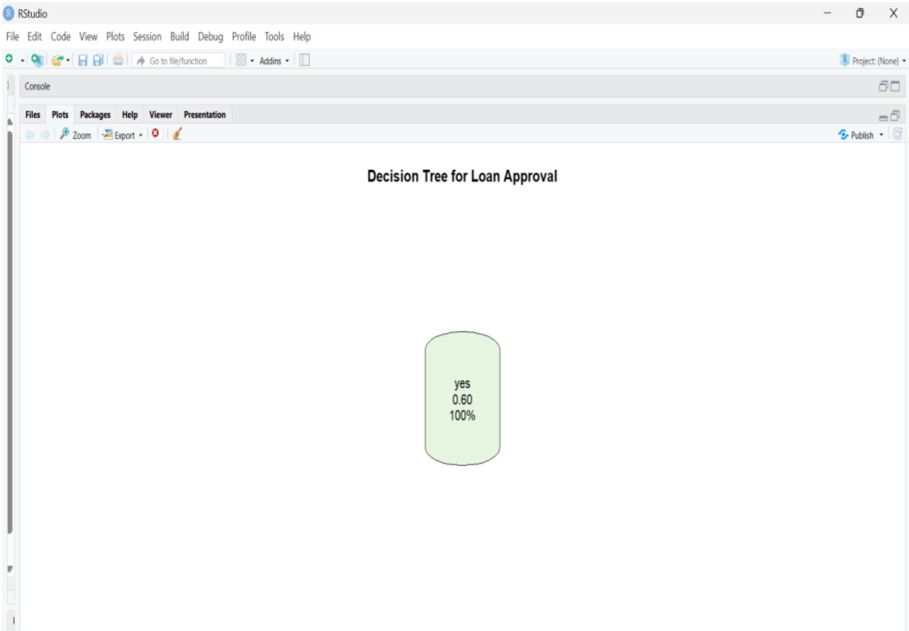
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no
young,medium,self-employed,average,yes
middle,low,employed,poor,no

ANS:
Decision Tree Classifier
The question asks you to use historical customer information to predict **Loan Approval (Yes/No)** using a Decision Tree. The dataset contains 15 records.

R Code:

```
data <- data.frame(
  Age = c("young","young","middle","old","old",
          "old","middle","young","young","old",
          "middle","middle","old","young","middle"),

  Income = c("high","high","high","medium","low",
             "low","low","medium","low","medium",
             "medium","high","medium","medium","low"),

  Employment = c("employed","self-employed","employed",
                 "employed","unemployed","self-employed",
                 "unemployed","employed","unemployed",
                 "self-employed","employed","self-employed",
                 "unemployed","self-employed","employed"),

  Credit = c("good","average","good","good","poor",
             "average","poor","average","poor","good",
             "average","good","poor","average","poor"),

  Loan = c("yes","yes","yes","yes","no",
           "no","no","yes","no","yes",
           "yes","yes","no","yes","no")
)

data[] <- lapply(data, factor)

model <- rpart(
  Loan ~ Age + Income + Employment + Credit,
  data = data,
  method = "class"
)

print(model)

rpart.plot(model,
          main = "Decision Tree for Loan Approval")
```
Output:





Result

A Decision Tree classifier was successfully constructed using **Age, Income, Employment Status and Credit Score** to predict **Loan Approval (Yes/No)**.
The tree can then be used to classify new customers based on their attributes.

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)
Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50
(i) Find which class had scored higher mean, median and range.
(ii) Plot above in boxplot and give the inferences

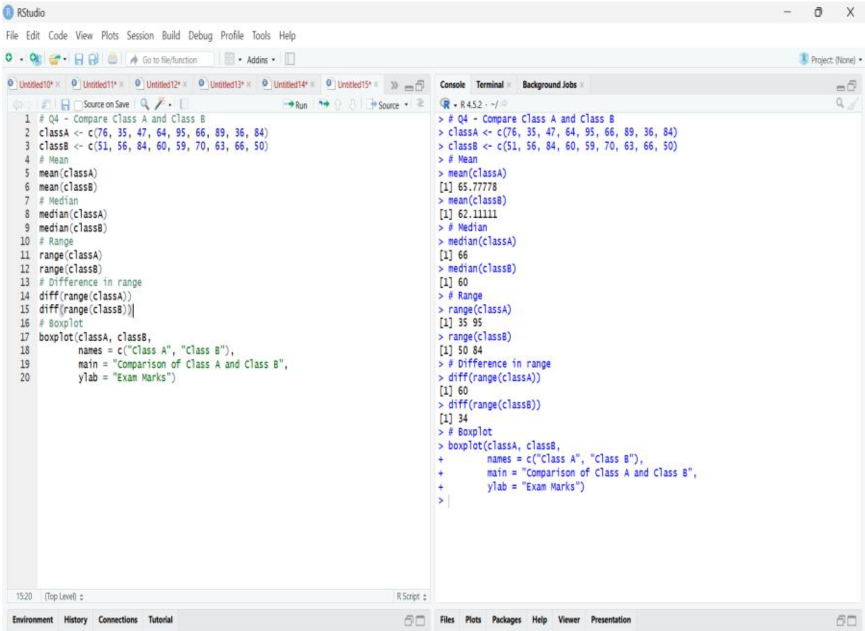
ANS:
Compare Class A and Class B
The assessment provides the following two Year 9 class results.
Data
Class A
76, 35, 47, 64, 95, 66, 89, 36, 84
Class B
51, 56, 84, 60, 59, 70, 63, 66, 50
R Code:
```
# Q4 - Compare Class A and Class B
classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
# Mean
mean(classA)
mean(classB)
# Median
median(classA)
median(classB)
# Range
range(classA)
range(classB)
# Difference in range
diff(range(classA))
diff(range(classB))
# Boxplot
boxplot(classA, classB,
        names = c("Class A", "Class B"),
        main = "Comparison of Class A and Class B",
        ylab = "Exam Marks")
```

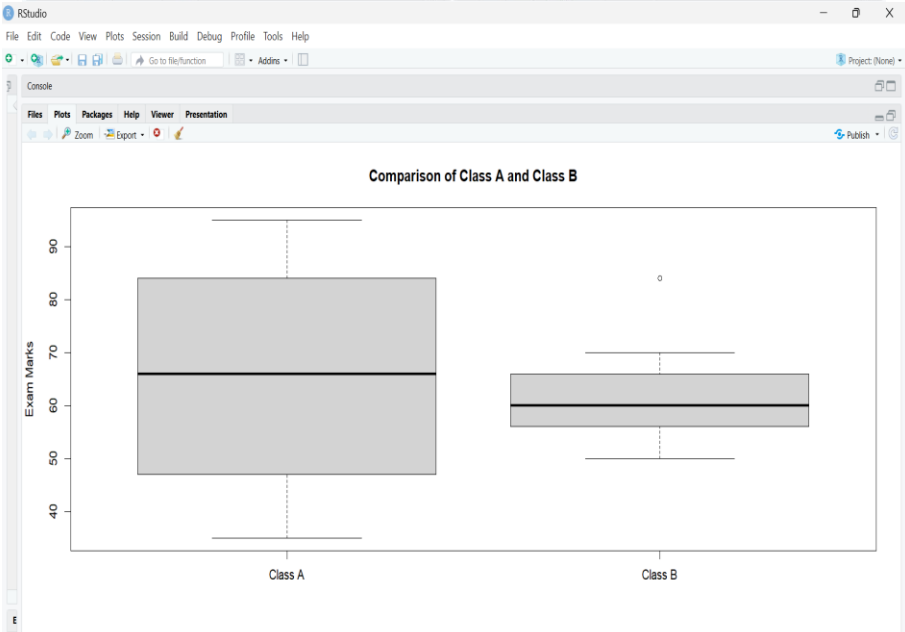
Results

Statistic	Class A	Class B
Mean	**65.78**	**62.11**
Median	**66**	**60**
Range	**60**	**34**

Result
Class A has:
- Higher mean → **65.78**
- Higher median → **66**
- Larger range → **60**

Output:





Inference

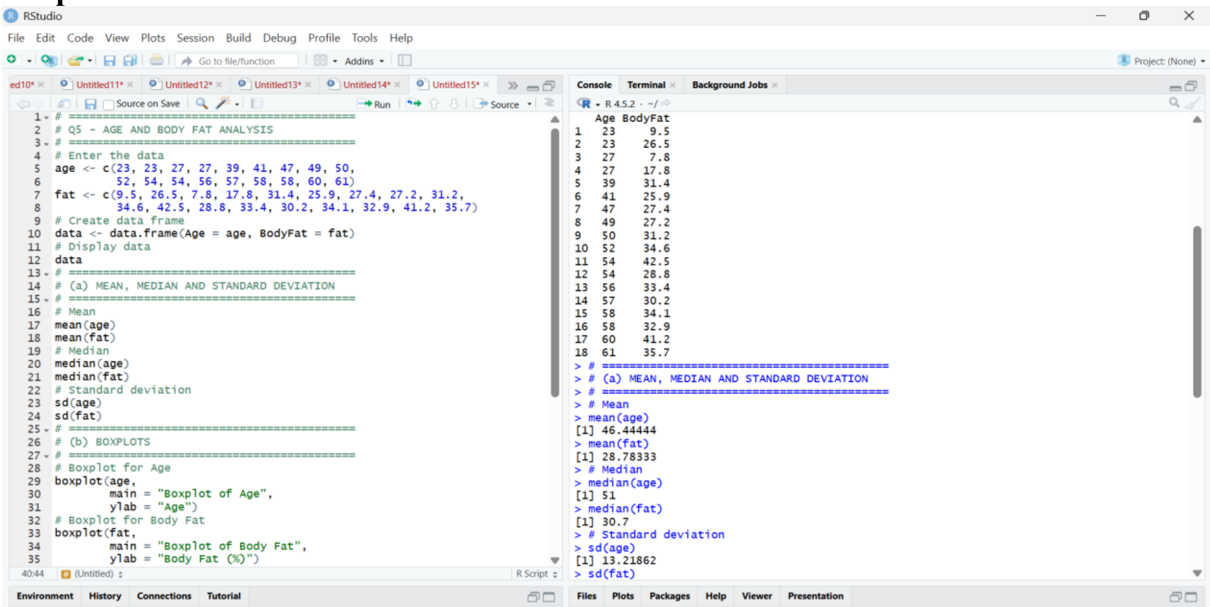
Class A performed better on average because it has a higher mean and median. However, Class A has greater variation in marks, as indicated by its larger range. Class B has more consistent performance.

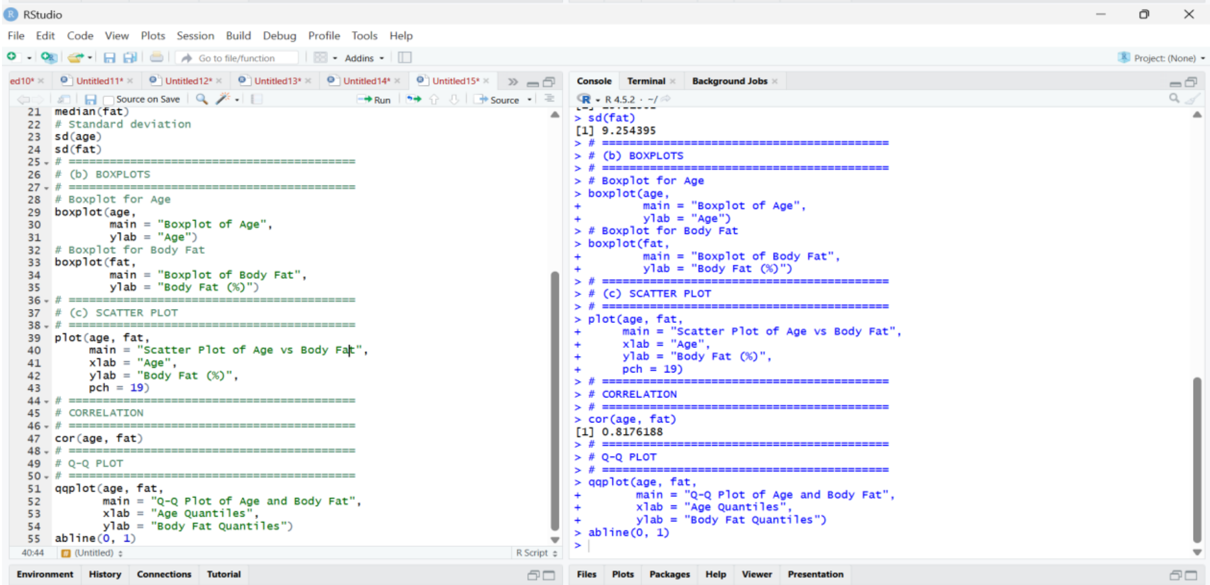
5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

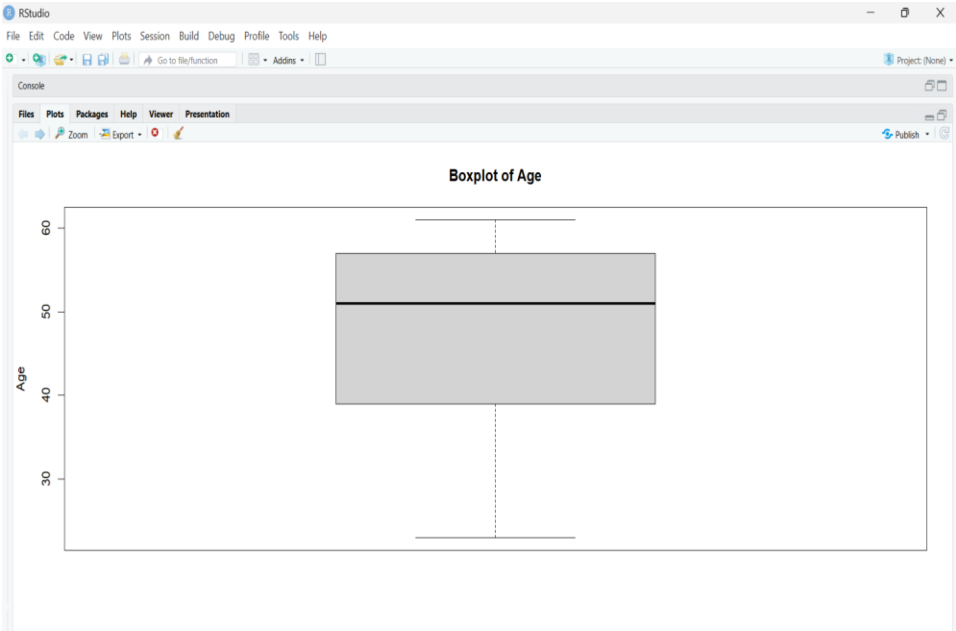
age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
(b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Output:



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Boxplot of Age
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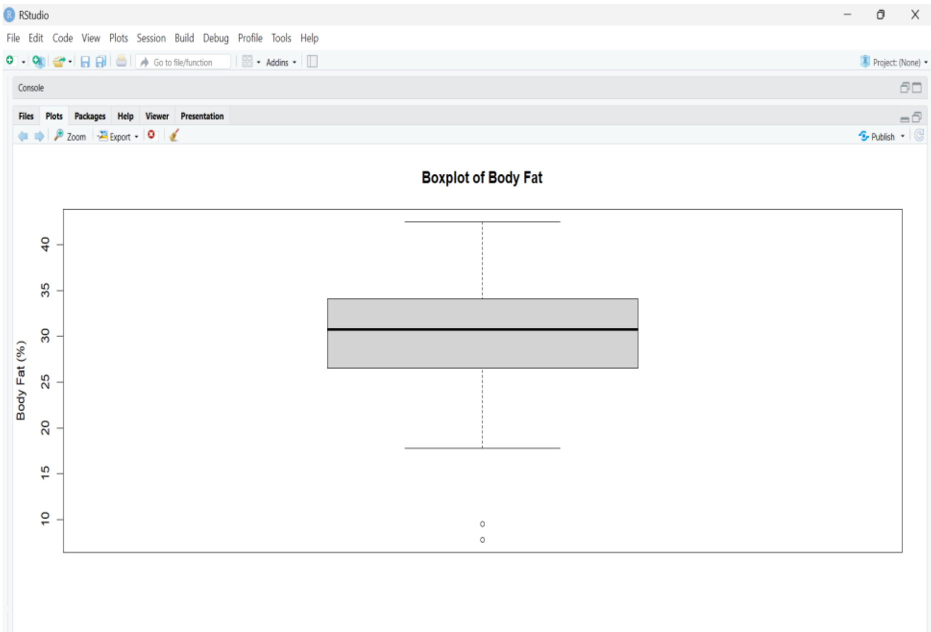
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Boxplot of Body Fat
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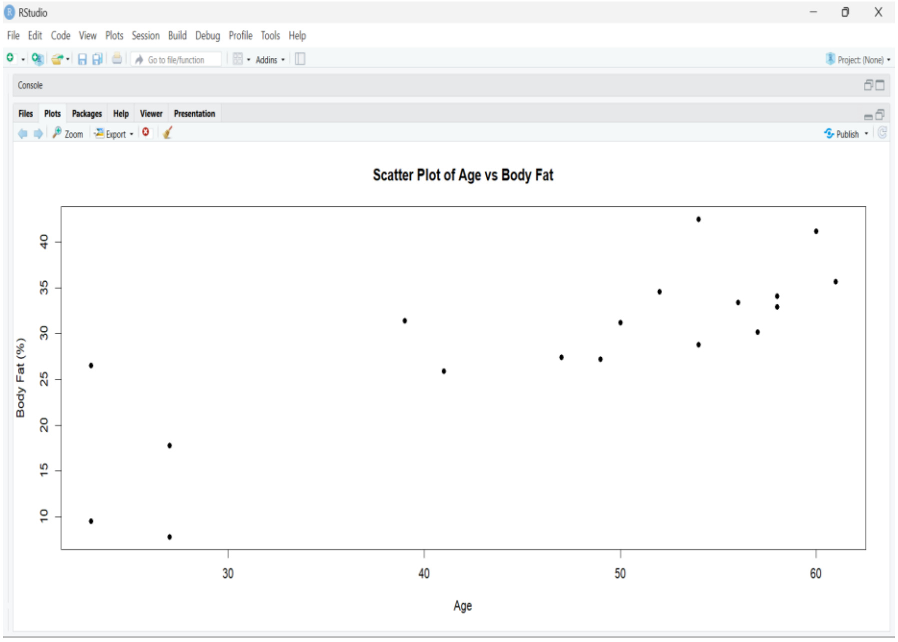
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Scatter Plot of Age vs Body Fat
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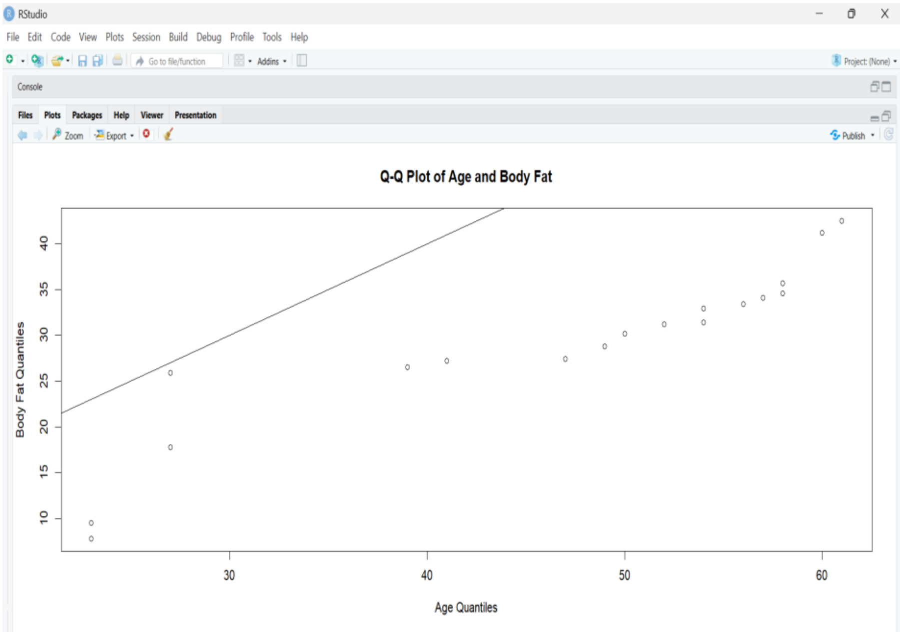
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Q-Q Plot of Age and Body Fat
Body Fat Quantiles
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Age Quantiles